The Upper Rhine Mobile Robotic Challenge

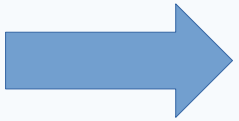
URMRC 26

The Upper Rhine Mobile Robotic Challenge

2026 "The 3rd Edition"

https://urmrc.github.io/URMRC/

- When ? 17 & 18 November 2026
- Where ?
  - Salon BE 5.0 – Industries du Futur, Mulhouse, France
  - More than 4453 visitors!





https://www.industriesdufutur.eu/en

- Who ?

Participants : 6 teams registered (ENSISA Mulhouse, CESI Strasbourg, TPS Strasbourg, HKA Karlsruhe (*2) and IUT Mulhouse) & one team currently in the process of registering.

  - 22 participants
  - Each team has its own dedicated mentor from its institution
  - Organizing team : Rodolfo Orjuela, Benoit Vigne, Jonathan Ledy.

- How ?
  - Two working days at the Salon BE 5.0
  - Four missions
  - Three winning teams

The Upper Rhine Mobile Robotic Challenge

Some pictures of the previous editions

November 2024 Mulhouse







November 2025 Mulhouse







https://www.youtube.com/watch?v=woctjWP4iLo
https://www.youtube.com/watch?v=raeM7qZx4qQ

The Upper Rhine Mobile Robotic Challenge

Hardware ? ==> TurtleBot3

Mobile robots will be used (perception and control):

- 2D camera, 360° LiDaR, IMU
- DYNAMIXEL motors x2
- Raspberry OpenCR board
- GROVE Electromagnet module
- Li-Po battery



Wi Fi

Raspberry Pi



The Upper Rhine Mobile Robotic Challenge

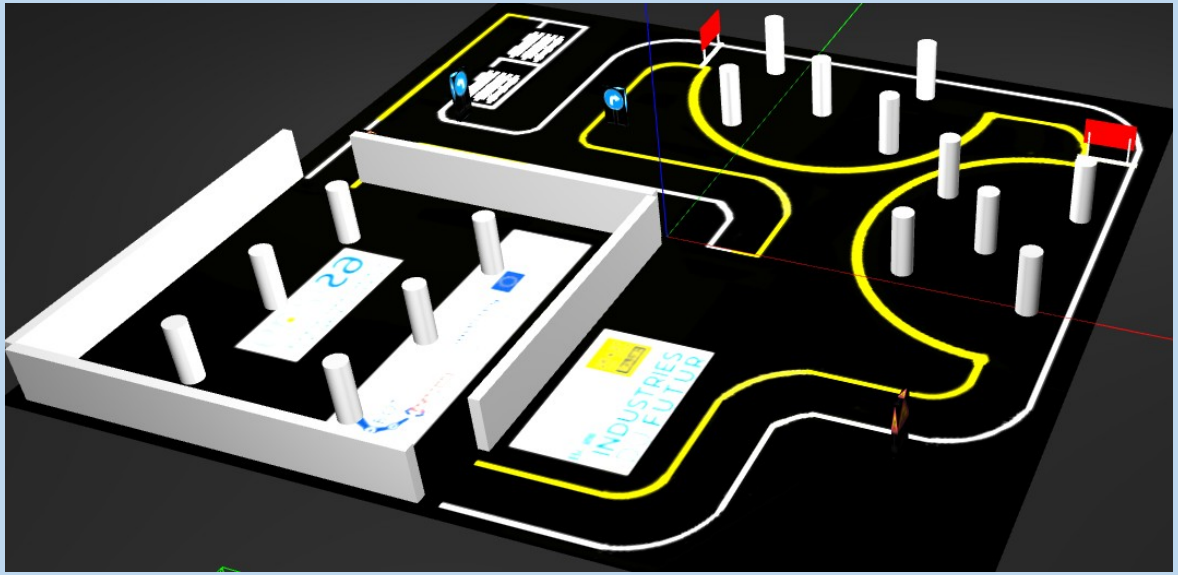
Software ?

ROS2

Gazebo environment

Image processing (OPENCV or Neural Network)

Navigation Stack



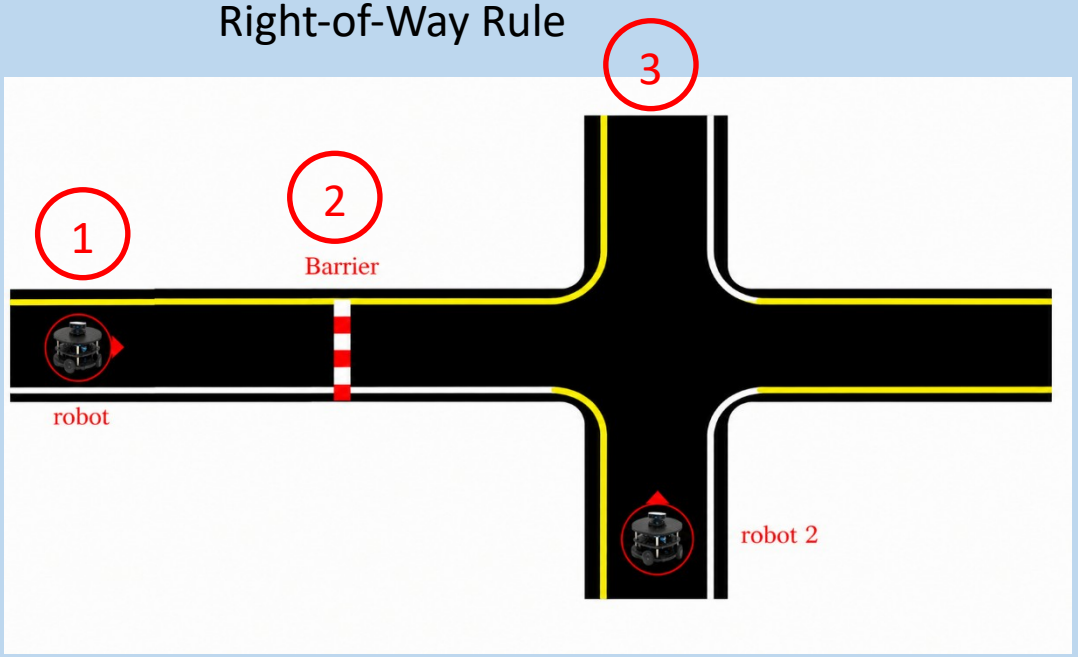
The Upper Rhine Mobile Robotic Challenge 2026

Missions

Autonomous navigation of mobile robots

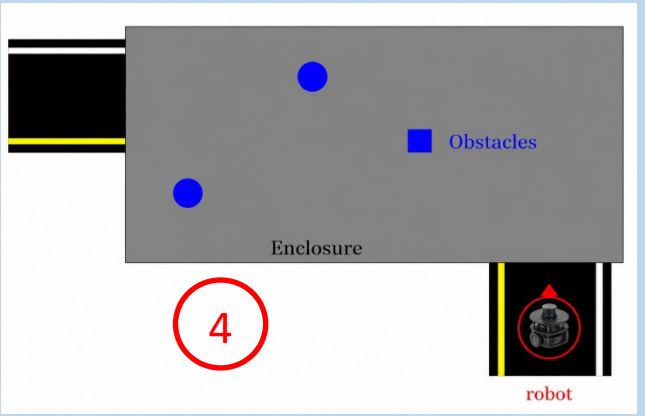
Track following missions

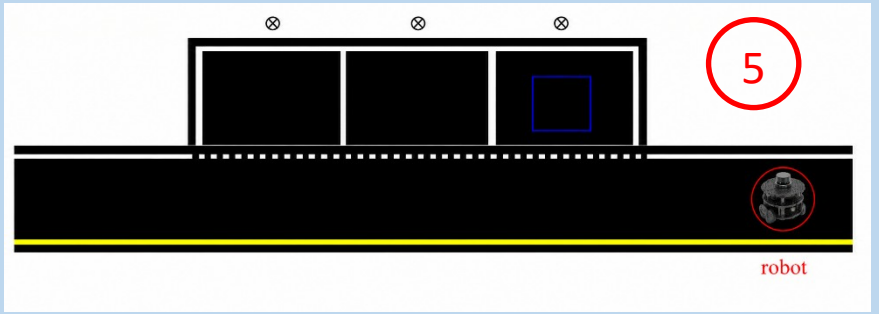
1. Track Following
2. Track Following with barrier detection
3. Track Following with an intersection & Right-of-Way Rule



Missions Related to Obstacle Avoidance and Maneuvering

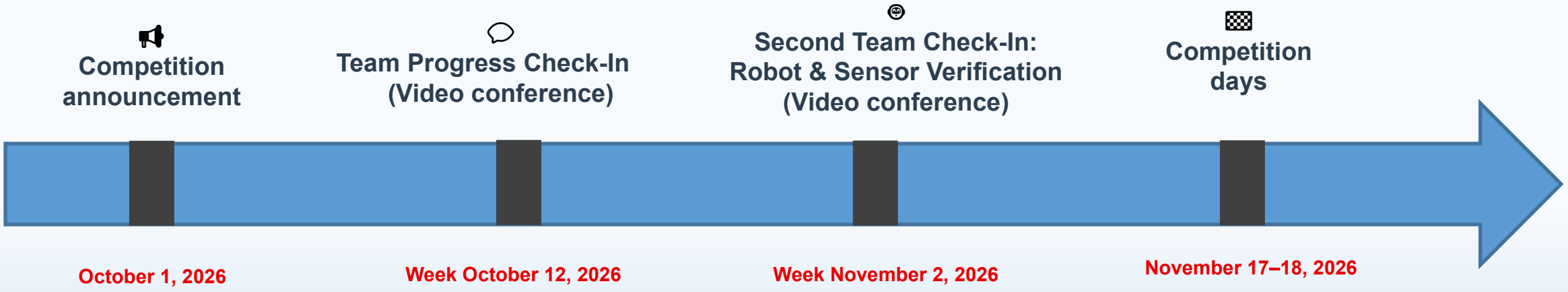
4. Navigating and Exiting an Enclosure
5. Storage and Retrieval Maneuver





The Upper Rhine Mobile Robotic Challenge

Schedule



The Upper Rhine Mobile Robotic Challenge

Competition days : Schedule

📅 **Day 1** — 9:00 AM to 6:00 PM
1-hour lunch break

- Three missions of Track-Following
- Collective Challenge

📅 **Day 2** — 9:00 AM to 3:00 PM
1-hour lunch break

- Obstacle Avoidance & Maneuvering Missions
- Collective Challenge
- 🏆 **Awards Ceremony — after 3:00 PM**

The Upper Rhine Mobile Robotic Challenge

- **Accommodation**

Participants will be accommodated at the youth hostel on the night of 17 November.

- **T-shirt Size Survey**

Each participant should select its size : https://evento.renater.fr/survey/t-shirt-size-3c7p39e1

The Upper Rhine Mobile Robotic Challenge

Welcome !!!